I-CoT: Implicit Chain-of-Thought Reasoning via Visual Cue Injection

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Abstract

We present a small pilot study investigating whether physical reasoning capabilities latent in Vision-Language Models (VLMs) can be unlocked through minimal prompt-level intervention. Motivated by the observation that human physical reasoning is guided by low-level perceptual heuristics—e.g., inferring motion direction from edge blur—we propose *Implicit Chain-of-Thought* (I-CoT): appending a short list of task-relevant visual attention cues to the question prompt, without any model weight updates. On a stratified 50-sample subset of PhysBench Chow et al. [2025] with Qwen2.5-VL-3B-Instruct Bai et al. [2025], both the baseline and I-CoT achieve 38.0% accuracy (19/50), with symmetric wins on the 12 contested samples. This null result is informative: it suggests latent physical knowledge exists in the model but that naive cue injection alone is insufficient to unlock it reliably. Code available at: `https://github.com/chazuz/ICoT`.

1 Motivation and Approach

Humans perform well on PhysBench Chow et al. [2025] not because they simulate physics, but because they apply fast perceptual heuristics: given an image of a moving object, a human immediately notices which edge is sharp and which is blurred, and concludes the object moves toward the sharp edge. No explicit reasoning is required—the correct image region is simply attended to.

This raises a practical question for VLMs: *is the necessary physical knowledge already present in the model, and can it be activated by steering the vision encoder's attention toward the right cues?* Rather than pursuing expensive fine-tuning, we test the cheapest possible intervention—prompt-level cue injection—as a decidable lower bound.

I-CoT. The baseline prompt wraps each question with *"Answer with exactly one letter: A, B, C, or D."* The I-CoT prompt is identical except that four perceptual concepts are inserted on a new line directly after the question sentence and before the answer choices. For example, a manipulation-sequence question receives `hand position, lid orientation, contact surface, object displacement`; a motion-direction question receives `vanishing point, perspective lines, spatial continuity, foreground depth`. No reasoning output is requested; the cues are implicit context only.

Practical note. We originally planned to hand-craft physics-grounded prompt templates. Under CPU-only constraints and limited time, we fell back on generating cues with Claude Sonnet 4.6 Anthropic [2025]—a methodological limitation we discuss below.

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Table 1: Contested samples: cases where BASE and ICoT predictions differ. ✓ marks the correct prediction. ICoT appends comma-separated visual attention cues directly after the question sentence.

idx	Sub-type	GT	BASE	ICoT	Winner	ICoT visual cues
8852	attribute	C	C✓	A	BASE	surface texture, deformation cues, material rigidity, edge sharpness
3914	manipulation	D	B	D✓	ICoT	hand position, lid orientation, contact surface, object displacement
3931	manipulation	B	A	B✓	ICoT	pen position, hand grip, bowl tilt, object clearance
1964	motion	C	A	C✓	ICoT	vanishing point, perspective lines, spatial continuity, foreground depth
2555	motion	A	A✓	C	BASE	vanishing point, perspective lines, spatial continuity, foreground depth
2431	motion	D	A	D✓	ICoT	vanishing point, perspective lines, spatial continuity, foreground depth
3983	manipulation	D	A	D✓	ICoT	flap position, fold crease, hand pressure, box gap
6779	manipulation	C	C✓	D	BASE	sock location, mango position, drawer state, hand displacement
2552	motion	C	A	C✓	ICoT	vanishing point, perspective lines, spatial continuity, foreground depth
3598	manipulation	A	A✓	B	BASE	tine sharpness, fork curvature, nail gap, leverage point
3442	manipulation	C	C✓	A	BASE	tine sharpness, fork curvature, nail gap, leverage point
2394	motion	A	A✓	D	BASE	vanishing point, perspective lines, spatial continuity, foreground depth

2 Experimental Setup

We evaluate on a 50-sample stratified subset of PhysBench (dynamics 50%, relationships 30%, property 16%, scene 4%), selected to maximise diagnostic coverage of task types where perceptual heuristics are most applicable. Qwen2.5-VL-3B-Instruct runs on CPU; images are resized to at most 200,000 pixels to prevent memory overflow on multi-image samples.

3 Results

Both strategies achieve **38.0% accuracy** (19/50). On the 12 contested samples (where exactly one strategy is correct), wins split evenly 6–6. I-CoT shows a modest positive signal on *relationships* (+1, 46.7% vs. 40.0%) and a small negative on *property* (-1, 37.5% vs. 50.0%).

Table 1 details the contested samples. For manipulation tasks I-CoT wins 5 of 6 contested cases, suggesting spatial cues help with step-sequence reasoning. For motion tasks, losses are concentrated on samples sharing identical generic cues (`vanishing point, ...`), pointing to over-generic cues as a distractor rather than a guide.

4 Limitations and Future Directions

Limitations. Cue generation via Claude Sonnet 4.6 introduces circularity: the intervention quality depends on a much larger external model's reasoning. The 50-sample CPU-only setup is statistically underpowered; a 6–6 split on 12 contested samples cannot distinguish a true null from sampling noise.

If pursued further. The symmetric disagreement in Table 1 suggests two tractable follow-up questions. First, attention map analysis could directly test whether cue injection shifts encoder attention toward task-relevant image regions. Second, the motion losses suggest that over-generic cues may distract rather than guide—lightweight token-weighting or prefix-tuning could give specific cues higher relative weight without full fine-tuning. More broadly, I-CoT establishes a near-zero-cost baseline point on the cost–performance curve that a thesis could use to fairly compare against CoT fine-tuning and LoRA Hu et al. [2022]. It would further be interesting to as well investigate the impact of varying numbers of cues as well as trying to quantify the influence of different cue qualities.

Scope of this work. This experiment is intentionally a pipeline demonstration rather than a research contribution. We originally intended a different direction of investigation e.g. measuring the impact of paraphrasing variations Schlegel et al. [2026] (e.g. impact of prompting from different perspectives). Its purpose is to show that I can independently design an experiment, implement a clean end-to-end pipeline under resource constraints, and reason critically about the results I hope to apply to the full thesis proposed.

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